
PROFILE

I have 5.5+ years of work experience in Data Science and Computer Vision with a proven ability and history of developing full-stack computer vision and data science projects. Hands-on experience leveraging Artificial Intelligence, machine learning, deep learning, and transfer learning models to solve challenging business problems.

PROFESSIONAL EXPERIENCE

Technical Consultant, ELK Educational Consultant

Oct 2021 – present
Chennai, India

1. A HIGHLY SECURE COMBINATION OF AUDIO AND VIDEO STEGANOGRAPHY USING HYBRID CRYPTOGRAPHY AND STEGANOGRAPHY

- In the project, My responsibilities included hiding the secret large text file in audio and video using Matlab.
- To achieve this, Initially, the input video undergoes processing in Matlab, where audio and video components are separated. Video frames are then extracted and enhanced using Gaussian and Median filters. Concurrently, secret data, stored in text files, undergoes Improved Huffman Encoding for compression and conversion. Further optimization involves binary-to-hexadecimal conversion, followed by Hex to ISO-8896 conversion, enhancing security and reducing data length. Compressed and converted data is concealed within video frames using the LSB method. AES-128 encryption is then applied to the combined secret data, ensuring confidentiality. Additionally, RSA encryption further secures the symmetric AES key, augmenting security through asymmetric cryptography. Finally, the combined audio and video, containing encrypted information, create the stego video/audio file.
 - Software details:
Back-end Process : MatLab
Used libraries: Image and Signal Processing Toolbox, Statistics and Machine Learning Toolbox
Front-end Process and Deployment: GUI and Matlab app

2. ALZHEIMER'S DISEASE CLASSIFICATION USING DEEP LEARNING

- For this project, I spearheaded the collection of a comprehensive dataset sourced from public repositories like Kaggle. Leveraging my leadership skills, I ensured the acquisition of a diverse dataset crucial for our project. Then, I converted the NII format images to visible PNG format, facilitating easier processing and analysis. Additionally, utilizing advanced preprocessing techniques, we enhanced and cropped the disease area of the image, guaranteeing optimal input for our model. The dataset was meticulously curated to encompass three distinct classes: CN (Cognitive Normal), MCI (Mild Cognitive Impairment), and AD (Alzheimer's Disease), ensuring comprehensive coverage of cognitive states.
- To tackle the classification task, we devised and executed a ResNet-18 model exhibited robust performance in accurately classifying cognitive states based on the provided image data. After training the model on the prepared dataset, we stored the learned weights for future use. Subsequently, we seamlessly deployed the model leveraging Docker for containerization and the Heroku platform for hosting.
 - Software details:
Back-end Process : Python and Matlab
Used libraries: Opencv, Keras, TensorFlow, Pandas, Numpy, Flask, Image Processing Toolbox

3. GAS LEAKAGE DETECTION AND ALERTING SYSTEM USING MACHINE LEARNING

- This project aimed to leverage machine learning techniques for the real-time detection of gas leakage in industrial settings. Our approach began with the collection of a substantial volume of structured sensor data tailored to our specific requirements. The sensors used included smoke sensors, LPG sensors, flame sensors, CO sensors, and CNG sensors, allowing for comprehensive monitoring of potential gas leaks. Emphasizing data quality, we meticulously cleaned and prepared the dataset to ensure it met the highest standards for subsequent analysis. Employing a variety of exploratory data analysis techniques, we delved deep into the dataset, uncovering valuable insights and discerning meaningful patterns crucial for informed decision-making. Additionally, we prioritized feature selection and engineering, utilizing advanced methodologies such as Principal Component Analysis (PCA) and Whale Optimization Algorithm (WOA) to enhance the discriminative power of our models. These techniques enabled us to extract the most relevant features, optimizing our model's accuracy and performance.
- For the classification task, we adopted the Random Forest algorithm, renowned for its robustness and efficacy in handling complex classification problems. By harnessing the power of Random Forest, we developed a highly accurate and reliable model capable of effectively identifying and classifying instances of gas leakage within industrial environments. Through our data-driven approach and strategic utilization of machine learning techniques, we achieved our primary objective of enhancing gas leakage detection in real-time industrial settings, contributing to improved safety and operational efficiency.
 - Software details:
Back-end Process : Matlab
Used libraries: MATLAB toolboxes

Front-end Process and Deployment: GUI GitHub+Matlab app

4. VIDEO SYNOPSIS AND VIDEO QUERY RETRIEVAL USING DEEP LEARNING

- In this project, we initiated by reading the input video file, laying the foundation for subsequent processing steps. We employed YOLOv5, a state-of-the-art object detection model, to detect objects within the video frames accurately. Leveraging deepSORT, an advanced object tracking algorithm, we tracked the detected objects across frames, ensuring continuous monitoring and analysis. Additionally, we utilized IWOA (Improved Whale Optimization Algorithm) for video synopsis, condensing lengthy video footage into concise summaries while retaining key events and activities. The integration of these techniques facilitated efficient and effective video analysis, enabling rapid understanding of video content. Furthermore, we employed Darknet-53, a powerful convolutional neural network architecture, for video query retrieval, enabling users to search for specific video segments based on predefined criteria. This comprehensive approach to video processing and retrieval streamlined the workflow and enhanced the usability of the system, contributing significantly to its success.
- My role involved accurately capturing and analyzing video content for the video synopsis process and query retrieval, contributing to the development of a robust video synopsis system.
 - Software details:
Back-end Process and Deployment : Python and Heroku
Used libraries: Sklearn, Pandas, Numpy ,OpenCV,TensorFlow,Matplotlib and Seaborn

5.E-HEALTHCARE RECORDS MANAGEMENT SYSTEM FOR PRIVACY PRESERVATION IN THIRD PARTY CLOUD PLATFORMS

- In this Project , as the back-end and front-end developer, my responsibilities involved Develop a highly secure E-Healthcare Records Management System to address privacy and security concerns related to personal data storage on public platforms.
- I Combining cryptography (AES) and steganography (LSB) techniques to ensure secure transmission and storage of healthcare records and I Utilized the Advanced Encryption Standard for encrypting and decrypting sensitive healthcare records, ensuring unauthorized parties cannot easily access the information.So I Employed the Least Significant Bit technique to hide the encrypted healthcare records within images, maintaining the visual quality of the images while concealing the data.Then I Utilized MySQL for cloud storage, providing a secure and scalable platform for storing encrypted healthcare records.
- The development environment was Windows, and the back-end processing was primarily conducted using Python, with key libraries including Flask, Tkinter, NumPy, PIL, and MySQL Connector,ensuring robustness and efficiency throughout the project lifecycle.

Software Developer, Spiro Prime tech services Pvt Ltd

2019 – 2021
Chennai, India

- **Multi-biometric System using AI**

- In my role as a AI Developer, I spearheaded the creation of an Multi biometric system using Image processing Techniques.This system aimed to integrate various biometric modalities, including fingerprint images, finger vein images, iris images, and face images, to enhance authentication accuracy and security. I Apply preprocessing techniques to enhance the quality and consistency of the captured images then I Utilize different machine learning feature extraction techniques for each biometric modality.
- I Design and implement a Convolutional Neural Network (CNN) for Training and Store the learned weights of the trained CNN for the System Recognition.

➤ Software details:

Back-end Process :Python

Used libraries: OpenCV, NumPy, SciPy,Scikit-learn,Tensorflow,Pandas

Front-end Process: GUI

 EDUCATION

Bachelor Of Engineering in Electronic and Communication Engineering,

2009 – 2013

Scad college of Engineering and Technology ,Cheranmahadevi.

Tirunelveli,

▪ Cumulative GPA : 7.5/10.0

Tamilnadu,India.

▪ Relevant Coursework : Digital Signal processing,Digital Image Processing,Embedded System

School Education,

Government Higher Secondary School, Vellalur,Madurai.

▪ HSC Cumulative GPA : 8.3/10.0

▪ SSLC Cumulative GPA: 8.3/10.0

 CERTIFICATES

- ♦ Data Analysis using AI

 SKILLS

Artificial Intelligence | Machine Learning | Deep Learning | Image

Processing | Computer Vision | Signal Processing| Data Analysis | EDA |

Python | Matlab | Flask Frame Work | Tkinter GUI Framework | Docker |

Github Process | Heroku |

Data Science Pipeline ((collection, cleansing, visualizaion, modeling, inierpreaiion), Sialisiics,
Hypothesis Tesing)

 INTERESTS

Traveling | Movies